Introduction to Data Structures

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Definition

- Data structure is representation of the logical relationship existing between individual elements of data.
- In other words, a data structure is a way of organizing all data items that considers not only the elements stored but also their relationship to each other.

Introduction

- Data structure affects the design of both structural & functional aspects of a program.

Program=algorithm + Data Structure

- You know that a algorithm is a step by step procedure to solve a particular function.

Introduction

- That means, algorithm is a set of instruction written to carry out certain tasks & the data structure is the way of organizing the data with their logical relationship retained.
- To develop a program of an algorithm, we should select an appropriate data structure for that algorithm.
- Therefore algorithm and its associated data structures form a program.

Data Structures

- Prepares the students for (and is a prerequisite for) the more advanced material students will encounter in later courses.
- Cover well-known data structures such as dynamic arrays, linked lists, stacks, queues, tree and graphs.
- Implement data structures in C++

Need for Data Structures

- Data structures organize data $\Rightarrow$ more efficient programs.
- More powerful computers $\Rightarrow$ more complex applications.
- More complex applications demand more calculations.

Organizing Data

- Any organization for a collection of records that can be searched, processed in any order, or modified
- The choice of data structure and algorithm can make the difference between a program running in a few seconds or many days

Efficiency

- A solution is said to be efficient if it solves the problem within its resource constraints.
  - Space
  - Time
- The cost of a solution is the amount of resources that the solution consumes

Selecting a Data Structure

- Select a data structure as follows:

1. Analyze the problem to determine the resource constraints a solution must meet.
2. Determine the basic operations that must be supported. Quantify the resource constraints for each operation.
3. Select the data structure that best meets these requirements.

Some Questions to Ask

- Are all data inserted into the data structure at the beginning, or are insertions interspersed with other operations?
- Can data be deleted?
- Are all data processed in some well-defined order, or is random access allowed?

Data Structure Philosophy

- Each data structure has costs and benefits.
- Rarely is one data structure better than another in all situations.
- A data structure requires:
  - space for each data item it stores,
  - time to perform each basic operation,
  - programming effort.

Goals of this Course

1. Reinforce the concept that costs and benefits exist for every data structure.
2. Learn the commonly used data structures.
   - These form a programmer's basic data structure "toolkit."
3. Understand how to measure the cost of a data structure or program.
   - These techniques also allow you to judge the merits of new data structures that you or others might invent.

Definition of Data

- The collection of raw facts and figures is called Data.
  - The word raw means that the facts have not yet been processed.
  - Data is collected from different resources.
  - Data may be facts, text, graphics images, sound/audio and videos.
  - We can’t make decision on the basis of Data.
  - Data is not meaningful.
- E.g. Student submits a form in a college.
  - The Administration can’t decide on the basis of this data, that which student may be admitted.

Types of Data we see in Computer

- **Numeric Data:** It consists of numeric digits from 0 to 9
- **Alphabetic Data:** This type of data consists of alphabetic letters from A to Z, a to z and blank spaces.
- **Alphanumeric Data:** It consist of numeric digits (0 to 9) and letters (A to Z & a to z) and all the special characters like +,*,@, etc like 87%.
- **Image Data:** Data Includes charts, graphs, pictures and drawings.
- **Audio Data:** Audio Data includes music, speech, or any type of sound.
- **Video Data:** Video is a set of full-motion images played at a high speed.

Statistical Types of Data

- **Qualitative Data**
  - It's a categorical measurement expressed not in terms of numbers, but rather by means of a natural language description

    For example: favorite color = "blue"

    height = "tall"
  - These categories may be ordered, these are called ordinal variables
  - Categorical variables that judge size (small, medium, large, etc.) are ordinal variables
- **Quantitative Data**
  - It's a numerical measurement expressed not by means of a natural language description, but rather in terms of numbers.

    For example: molecule length = "450 nm"

    height = "1.8 m"
  - ***However, not all numbers are continuous and measurable E.g CNIC#.***

Types of Data Structure

Data structure are classified into two type such as linear or non-linear.

- **Linear:** A data structure is said to be linear if its elements form a sequence
  - It's either represented by means of sequential memory locations or the other way is to have a linear relationship between the elements represented by means of pointers or links.
  - E.g Array, Link List
- **Non-linear:** A data structure is said to be non-linear if its elements have a hierarchical relationship among them.
  - All elements assign the memory as random form and you can fetch data elements through random access
  - E.g such as trees and graphs.

Creating Data Structures

There are two approaches for creating a data structure.

Static data structure: the size of the structure is fixed

- It is very good for storing a well-defined number of data items
- a programmer might be coding an 'Undo' function where the last 10 user actions are kept in case they want to undo their actions

Dynamic data structure: Are used where the number of items to be stored is not known in advance

- It allows grow and shrink as the demand for storage arises
- For example a programmer coding a print spooler will have to maintain a data structure to store print jobs, but he cannot know in advance how many jobs there will be

Static Vs Dynamic Data Structure

Static

- Memory is allocated at compile time
- Advantages
  - There will be no problem with adding and removing data items
  - Easier to program as there is no need to check on data structure size at any point
- Disadvantages
  - It is very inefficient bcz wastage of memory if not used

Dynamic

- Memory is allocated dynamically at Execution Time
- Advantages
  - Makes the most efficient use of memory as it needs
- Disadvantages
  - Might 'overflow' or exceeds its limit or 'underflow' or become empty
  - Harder to program because of tracking data item at all times

Field

- A group of related characters, representing some unit information is called a Field
- Different Fields in a record can be Roll No., Student's Name, class and Student Address

Roll#	NAME	F/Name	Address
1501	Haris	Sameer	Peshawar
1505	Umair	Nasir	Charsadda

Record/Row/Tuple

Record/Row/Tuple

Record:

- A collection of related fields treated as a single unit is called a record
- E.g. a student record includes a set of fields that contain Roll No., Student's Name, class and Student Address

Roll#	NAME	F/Name	Address	
1501	Haris	Sameer	Peshawar	Record/Row/Tuple
1505	Umair	Nasir	Charsadda	Record/Row/Tuple

File:

- A named portion of secondary memory (magnetic tape or hard disk) that is allocated for the purpose of storing physical records is called a physical file
- A collection of related records treated as a single unit is called file
- A student file may contain the records of hundreds of students

Example Student File

Roll#	Name	Dept_Name	Salary	Course Title	Date Completed
100	Sabir Ullah	Web Designing	48,000	Adobe Photoshop	19/06/2000
100	Sabir Ullah	Web Designing	48,000	Dream Weaver	10/07/2000
140	Mansoor Noor	Networks	40,000	CCNA	12/09/2000
110	Zubair Khan	Information System	50,000	Visual Basic	21/05/2000
110	Zubair Khan	Information System	50,000	C++	25/08/2000
190	Yasir Nawaz	Web Designing	35,000		
150	Ijaz Khan	Information System	60,000	JAVA	24/02/2000
150	Ijaz Khan	Information System	60,000	MS Office	01/09/2000

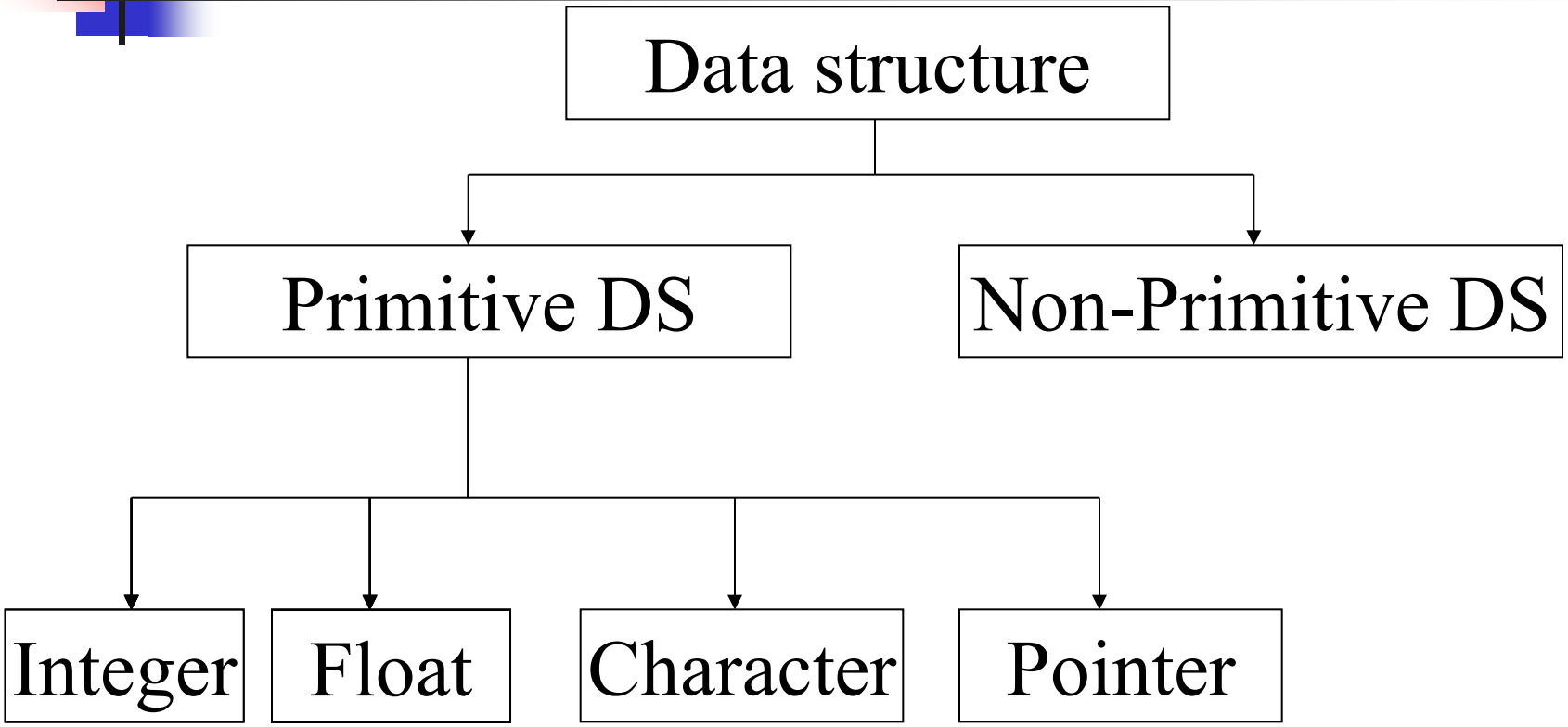
Definition of Information

- When data is processed in such a way that it increases the knowledge and makes him/her capable to take decision about the relevant situation is called information.
  - It is organized or processed form of data.
  - It more meaningful than data.
  - It helps us for making different decisions.
- E.g. Merit list of student.

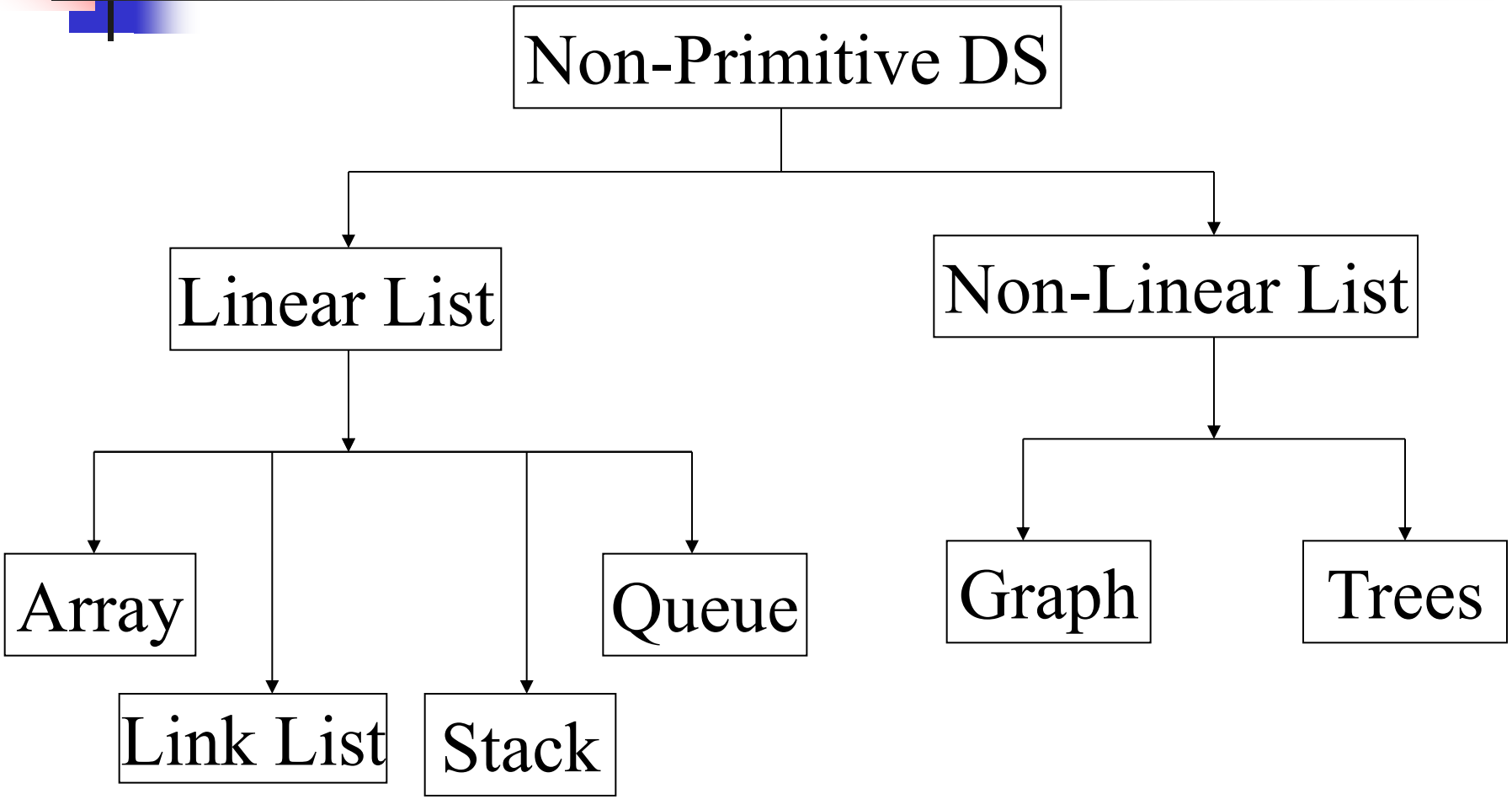
Classification of Data Structure

- Data structure are normally divided into two broad categories:
  - Primitive/Physical Data Structure
    - Also called Built-in Data Types
  - Non-Primitive/Logical Data Structure
    - Known as User Defined Data Types

Classification of Data Structure



Classification of Data Structure



Classification of Data Structure

Primitive/Physical

- Integer
- Float
- Character
- Pointers

Non-Primitive/Logical

- Non Linear
  - Graph
  - Tree
- Linear List
  - Stack
  - Queue
    - Dequeue
    - Circular Queue
  - Array
    - One Dimensional
    - Multi Dimensional
  - Link List
    - One Way/Singly Link List
    - Two Way/Doubly Link List
    - Circular Link List

Primitive Data Structure

- There are basic structures and directly operated upon by the machine instructions.
- In general, there are different representation on different computers.
- Integer, Floating-point number, Character constants, string constants, pointers etc, fall in this category.

Non-Primitive Data Structure

- There are more sophisticated data structures.
- These are derived from the primitive data structures.
- The non-primitive data structures emphasize on structuring of a group of homogeneous (same type) or heterogeneous (different type) data items.

Non-Primitive Data Structure

- Lists, Stack, Queue, Tree, Graph are example of non-primitive data structures.
- The design of an efficient data structure must take operations to be performed on the data structure.

Non-Primitive Data Structure

- The most commonly used operation on data structure are broadly categorized into following types:
  - **Create/Insert**
    - Adding a new record to the structure
  - **Destroy or Delete**
    - Removing the record from the structure.
  - **Traversing**
    - Accessing each records exactly once so that certain items in the record may be processed
  - **Searching**
    - Finding the location of a particular record with a given key value, or finding the location of all records which satisfy one or more conditions

Non-Primitive Data Structure…..

- **Selection**
  - Selecting a particular record with a given key value, or selecting all records which satisfy one or more conditions for other operations
- **Updating**
  - Updating an element at a particular position.
- **Sorting**
  - Managing the data or record in some logical order(Ascending or descending order)
- **Merging**
  - Combining the record in two different sorted files into a single sorted file

Different between them

- A primitive data structure is generally a basic structure that is usually built into the language, such as an integer, a float.
- A non-primitive data structure is built out of primitive data structures linked together in meaningful ways, such as a or a linked-list, binary search tree, AVL Tree, graph etc.